

2023

Flix2You

A Team Dominoe® Solution

IST 210
Section 002
Professor Aebli

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Table of Contents

Executive Summary.....	4
Project Management	6
Labor Budget.....	8
Server Upgrades.....	8
User Analysis	9
Users	9
Database Administrator	9
Information Technology.....	9
Management.....	9
Operations	9
Human Resources	10
Marketing.....	10
Customers	10
Organization of Data	11
Existing Database Analysis – Denormalization Approach	11
Changes made to Existing Flix2You ERD:	11
Database Administration	13
Backups and Recovery	13
Data Access and Security	14
Data Load	15

Executive Summary

1. Estimated Time Frame
 - a. The proposed project timeline is estimated to be **two months**. In this two-month time frame the contractors (Team Dominoe) plan to upgrade current system hardware, software systems, and implement employee training.
 - i. Please refer to Project Management for a deeper understanding.
2. Budget
 - a. Labor - \$97,800.00
 - b. Hardware & Software - \$11,016.95
3. Legal
 - a. Information provided from the legal department will capture and take care of virtual data collection, use of website cookies, and protect the Flix2You company.
 - i. Please refer to the Legal department to view the Privacy Statement.
4. Return on Investment (Revenue)
 - a. The proposed budget plans to help generate revenue through means of marketing, increasing the pace at which business transactions are processed, and increasing production from the current employee base.

Project Summary

Team Dominoe has examined the Flix2You database and is honored to introduce a new business model to the Board of Directors. We intend to utilize what has already been in place at Flix2You to forward the business. Team Dominoe understands the competition offering solutions, but we can assure the board that the team is ready to not only work efficiently and effectively in comparison to our competitors. By the end of our work, Flix2You will not only be data rich, but also information rich.

It is important to mention the benefits regarding cost and budgeting when looking at Flix2You. Team Dominoe has looked at labor costs, software costs, and external costs and has summed up a fair price for what Flix2You will receive as a business. Looking into the future, the return on investment given to the company will skyrocket over competitors such as Amazon, Netflix, or Hulu. Given how much we at Team Dominoe are charging, the return on investment will be the greatest possible outcome in comparison to our competitors.

The actual data being pulled from the newly designed software is to fully help export, which will allow for quicker data retrieval. The use of normalization and denormalization on the new system will help create future expansion for multi-app development. The use of the new techniques will also help increase security and safety of the storage.

In any business, the need to grow is what drives the bottom line. Team Dominoe has assessed the current situation at Flix2You and has a general game plan to increase your company's size, revenue, and efficiency. Many of the current hardware and software applications being used by the employees at Flix2You are bottlenecking the true capabilities of the company. By modernizing the company servers and updating the way data is utilized and thought about, we feel exponential growth is possible at Flix2You. Marketing has become a specific tool and we would like to help individualize every ad that is sent out, helping bring in every customer possible. Without further a due, let us jump into it!

Project Management

With our company, project management is taken seriously. We design a flexible yet sturdy time to meet our clients' criteria for success as well as the time required for our team to ensure that success. The task list below provides a visual timeline of what we believe the required steps are to increase revenue at Flix2you through a meticulous and well vetted process.

TASK AT HAND	TIME FRAME
Meet and Discuss Executive Level Needs	Day 1-3
Meet and Discuss Employee Level Needs	Day 4-5
Team Domino Meeting – Ensures development needs are understood	Day 6-9
Secure hardware and software upgrades for server	Day 10-17
Install new hardware and software upgrades to server	Day 18
Begin implementing new ERD for all data	Day 18-28
Begin implementing front end forms for employee use	Day 28-30
Begin website upgrades for data acquisition	Day 30-35
Produce specific queries to profile existing customers	Day 35-40
Compile and submit first wave of reports	Day 41
Implement any changes to increase productivity and report data.	Day 41-55

Days 1-5

Meeting and understanding client needs are very important to us at Dominoes Enterprise. Creating a list of deliverables to ensure your company's success starts at the human level. First, we must understand what the company is – our team will have an in depth understanding of all operations at your business by the time our executive level meetings end. Next, we meet with the employees to find out the best practice for implementing the new strategy in their work. This is where we find out what is most important for their productivity and your revenue. It remains of the utmost importance for our team to understand your team to help generate additional revenue.

Days 6-17

Our team will meet at our location and begin the barebones work. We take the information collected through the employee meetings and start to visualize what that looks like to us. This is where we design the framework for what will come and answer some of our own questions. At some stages this allows us to know that potential hardware and software upgrades may not match existing needs the clients have had to address before any purchases are made. This stage also allows us to deep dive into how the data should be stored, create our ERDs, and allow our team to start writing SQL that can be implemented when the servers are ready.

Day 18

A physical upgrade to your systems will be carried out. This transition will be made during non-peak hours and generally won't take longer than 12 hours. We do, however, reserve the entire day as sometimes data migration can be slowed from sources largely outside of our control. While this may seem like a long time in terms of hours, some full migrations, say from a physical to an AWS server, could take months of migration and work to get fully up and running. By utilizing current servers with upgrades, we chop a lot of that time off.

Days 19-35

Our team has taken the required time to create our framework. The implementation of the new ideas collected from the executive level meetings and employee level meetings start getting put into play. This is where our team begins to overdrive, our website developers begin making necessary changes to the existing site to better collect and store incoming new data. Our data analysis experts are working on moving the existing data and formatting in a way that is useful for your current employees to pull. This is what we refer to as denormalization. We take the data and denormalize it. This allows for current employees to pull information as they need it. While this process can take more processing power, the necessary upgrades have been made to ensure no time is lost by your company.

Day 35 and Onward

This is where our team works side by side with data gathered or previously stored by Flix2you. We continue to produce reports and help feed sales information to you and your advertising department. At this stage, you should see targeted ads become easier to produce, thus cutting money from your current marketing budget and allowing you to expand your outreach by reinvesting or taking that revenue and utilizing it elsewhere. Dominoes Enterprise will work with your team to continually improve understanding of the new systems and create an efficient way to process all your data in terms of new revenue.

Our team never stops working for our clients. During the early stages we gain intimate knowledge of our clients' business operations. We budget in a way that provides windows of opportunity for both us and our clients. When the implementation of data is done, we remain available. We want to see our customers thrive off the innovation our team can help create, and with that comes a long-term contract to help to

maintain and utilize the data systems we have created for you. Know that in your success is also ours.

Labor Budget

JOB TITLE	EMPLOYEE NAME	HOURLY WAGE	EXPECTED HOURS
Project Management	Richard Warnken	\$65.50	400
Database Administrator & Quality Assurance	Gavan Quimby	\$79.40	400
Database Analyst	Gregory Harrington Jr.	\$38.00	200 (per job)
Programmers	Gregory Harrington Jr., Bhavya Patel	\$37.00 / \$51.00	400
Total			\$97,800.00

Server Upgrades

Component	Model	Cost	Qty.	Extended Cost
Server & CPU	GIGABYTE R282-X93 w/ AMD EPYC 7F72	\$4974.98	1	\$4,974.98
Software	Microsoft SQL Server 2022 – Standard Ed.	\$3945.00	1	\$3,945.00
Storage	SAMSUNG EVO 2TB SSD	\$259.99	1	\$259.00
	SAMSUNG EVO 4TB SSD	\$499.00	1	\$499.00
Operating System	Windows Server 2022 – Standard Ed.	\$1069.00	1	\$1,069.00
Memory	Crucial DDR4 2x16	\$68.99	1	\$68.99
RAID Controller	LSI MegaRAID Low Profile Controller	\$199.99	1	\$199.99
TOTAL				\$11,016.95

User Analysis

Users

The Flix2 You Company has several specific markets which it applies to, according to our analysis. It just makes sense to encourage every community of people or staff inside their organization to see what is appropriate. Which would ensure that all activities stay safe, with no risk that unauthorized visitors would be able to change the database. Since our intentions are to repair the existing database through standardization, we won't have specific tables at this point to which each category would grant access. Alternatively, if we are selected to take on this mission, we would offer general ideas which what team will be in control of. We've built a user category that lists and gives an overview of what they can access and update.

Database Administrator

At the very top of the pyramid is the database manager. There is only one administrator, preventing confusion from leading to more mistakes being added to the database. They can edit, view, and create tables because they have access to the entire database. They also have the power to grant access to other areas of the business, for example, determining which portions of the database the IT department is permitted to use for each job. The administrator will be able to handle any issues or worries the departments may be experiencing.

Information Technology

The day-to-day operations of the company will be managed by this team, which will also be expected to manage the database's daily operations. The database administrator can grant access to the database and assign sections of the database based on the work since this team would have a very large working role.

Management

This department, which will report directly to the database administrator, will have access to all database elements, including personal data, and can see and modify them.

Operations

While a customer may need assistance with shipping and transaction issues, the operations staff will be in charge of handling the transaction itself. They oversee shipping out the rentals and billing the customers. We have chosen to separate billing and shipping because they are two totally different processes. Only the information (such as addresses) of customers and the movies they wish to rent will be visible to those in charge of shipping. The customer queue, which shows the order in which each user has rented a movie, will also need to be kept track of. The billing department won't

be able to create tables, but it will be able to view and alter a customer's details, transactions, and accounts.

Human Resources

Human resources are required for both the protection of employee data in our database and the resolution of any issues that may be of concern to employees. This section of the company will be approved to view all employee data and modify it as necessary.

Marketing

All database sections related to clients and their purchasing activities must be viewed by the marketing team. The marketing team will be able to calculate reports using the information above and help to make decisions about any possible future activities the company may undertake. If selected for the job, a more precise illustration of the specific tables they would oversee would be made.

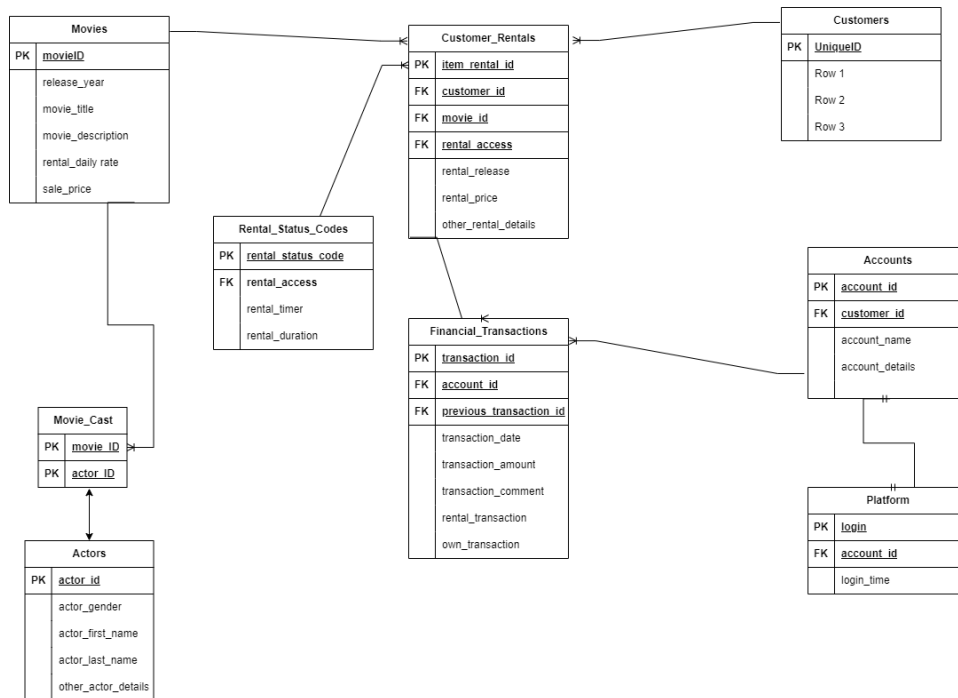
Customers

Customers are only allowed to browse, buy, or rent the videos that the company offers. They will also be allowed to create an account and to a certain extent customize it. Customers will have access to both videos that may be related to ones they have already rented as well as their rental history.

Organization of Data

Existing Database Analysis – Identifying Relationships

Figure # - New ERD



Existing Database Analysis – Denormalization Approach

With an ever-changing environment for the movie rentals business, the team was ready to develop results that could modernize the database into the present. Team Domino has decided to use denormalization by having the original database processed into our newly designed ERD.

In Other words, the original database is feeding its data into the newer and more simplistic database. The Updated ERD will then make use of the newly upgraded systems (check addendum for more information) when feeding the necessary data that the dashboard is requesting.

The biggest benefit of Flix2You programmers utilizing the denormalized dataset is being able to access the data with simpler SQL functions. With the normalized tables joined together, the admins will be able to export the data more efficiently and have it retrieved in a timely manner.

Changes made to Existing Flix2You ERD:

1. Since Flix2You is becoming more modernized with other streaming services, anything related to a physical rental store has been removed in the new ERD and replaced with the modernized approach of online streaming. This includes the removal of the Video_Stores table.
2. The Payment_Methods table is unnecessary since there isn't a physical location, and everything will be paid virtually. The Main change is the Payment_Methods table is removed and the payment_method_code attribute is removed in Accounts.
3. With the removal of Payment_Methods table, the Transaction_Types table also becomes obsolete. The Transaction_Types table is removed as well as the foreign key in the main Financial_Transaction table.
4. The Genre_Codes table feels overcomplicated. The admin who accesses the database won't need the genre_description attribute ever. The Genre_Codes table will be removed and the genre_type_codes attribute in the Movies table will no longer be a foreign key.
5. With Flix2You going digital, there isn't any need for the Format_Types table. The Format_Types table and foreign attributes have been removed.
6. Condition_Codes table has also been removed due to Flix2You going virtual and will no longer be offering DVDs.
7. Instead of thinking the Rental_Status_Codes is for physical rentals; the table will be reshaped to function for virtual rentals. Removed rental_status_description attribute and implemented rental_timer and rental_duration. Timer being how long the user has to access the rental and duration being how long the user can purchase the rental for (6 hours, 12 hours ,24 hours).
8. With combinations and reworked tables, the attributes in the beefier tables also need to be adjusted. In the Financial_Transactions table, the removed attributes are transaction_type_code and item_rental_id. The added attributes are rental_transaction or own_transaction for whether the user is buying or renting the movie on the platform.
9. Team Domino has also noticed that the Customer_Rentals table needs to be readjusted. Removed attributes include rental_date_out, rental_date_returned, rental_amount due and rental_status_code. The new repurposed attributes for the Customer_Rentals table includes rental_release, rental_access and rental_price. Release will mark when the rentals are available to purchase, access will be a FK from Rental_Status_Codes that indicates whether or not the user has access to the rental, and price is the price of the specified rental.
10. The last change to the overhaul is the addition of a Platform table which tracks login and account data. This table is connected to the Accounts table in a one-to-one relationship. This is because one account should only have one account login.

Database Administration

The Flix2You database administrator's role and responsibilities include establishing policies and procedures that would be used in protecting and controlling the database systems.

Elaborating on a database administrator's role and responsibilities, their duties would include controlling the database's structure, managing changes to the database's data, and routine maintenance on system hardware and its software and applications. Flix2You's database administrator would also have the ability to delegate rights and privileges to individuals, routinely members of the Flix2You company or otherwise related, who require access to Flix2You's computer systems.

Certain tasks of database administration may be offloaded to fellow members of Flix2You's IT Department as deemed necessary.

Backups and Recovery

Flix2You should have their data stored in a relational database management system; specifically with our solution, stored with Microsoft SQL Server. With Microsoft SQL Server, Flix2You would be able to store their data on multiple database objects, most common being database tables, in which are stored in multiple storage devices (Solid State Drives and/or Hard Disk Drives). There are plenty of benefits to utilizing multiple storage devices rather than only one storage device including

Microsoft SQL Server offers four different methods for data backup and recovery:

- Full Backup
- File Backup
- Transaction Log Backup
- Differential Backup

Understanding of Terminology: Between these four methods of backup and recovery, Full Backup and File Backup both pertain to backing up the Flix2You database itself whilst Transaction Log Backup and Differential Backup both pertain to backing up changes that were made to the files within the Flix2You database.

A **Full Backup** allows for a backup of the entire Flix2You database. With a Full Backup, the database will still be operational and able to be accessed during the backup process. A snapshot of the transaction log of the database will be included with the backup to ensure continuity of data in the database during the backup and the data in the resulting backup file(s).

Full backups are integral to any backup plan and should be done at least once. It is also important to do Full Backups on a daily basis at nighttime to prevent issues that may

arise with multiple users of the database. Full Backups should be the first solution that businesses look to for backing up data, but in cases of businesses with a massive database, a Full Backup would take a long time to complete and may not be the most desirable solution in this case.

When data recovery is required using a Full Backup, Microsoft SQL Server would essentially regenerate all of the required files for the database at the time the backup was completed.

A **File Backup** is to be used when a database is too large for a Full Backup to backup within a reasonable amount of time. A File Backup has potentially many upsides versus a Full Backup, that could include quicker backup and recovery times. A File Backup should be used in a case where say a couple of files have been corrupted or lost, so that just those couple of files would simply be recovered.

It is crucial to keep note that a File Backup and Recovery cannot proceed on its own. To keep consistency for the entire database when File Recovery is used, elaborating, ensuring that the representation of time in the files match up, the transaction logs for the database must be consulted.

A **Transaction Log Backup** will create a copy of the transaction log, which keeps track of changes made to the database. Once a Transaction Log Backup is completed, all document changes to the database will be removed from the transaction log. With our plan, we would utilize a Full Backup daily at nighttime and utilize a Transaction Log Backup for every transaction that would take place.

To recover a Transaction Log, you would need to go to the last full backup of the database and obtain every transaction log since that full backup.

A **Differential Backup** will store all changes made to the database since the last backup. Using a Differential Backup would create a copy of all changes made to the database since the last full backup, akin to a transaction log.

Our backup plan would include performing a Full Backup on the server daily at nighttime, and we would perform a Transaction Log Backup every time a transaction would occur on the server.

With our backup plan, should a critical situation arise, all that would be needed to be done is to use the full backup to restore the database, and use the transaction log backup to restore the transaction log.

Data Access and Security

Flix2You should have a hierarchy of its users in order to keep its database secure from threats from both within and outside the company. Given that Flix2You is an online streaming service that would obtain customers' sensitive information (credit cards, names, dates of birth, etc.), we will have a heightened focus on the servers' security.

Within this hierarchy of users, only the database administrator and the other IT employees would have access to the database. The database administrator would have the most access; with responsibilities including providing access to update the database and scheduling antivirus scans and software updates. Flix2You's IT team will be the only employees that are given the privilege to update the database.

We will also utilize biometric scanning in the form of fingerprint scanners for Flix2You's data room in order to physically protect Flix2You's server. When it comes to virtual protection, our lead programmers are leading red team-blue team exercises to detect and patch any exploits in Flix2You's server before they can be abused.

Our privacy statement should include disclaimers that we are collecting and using their data in order to find out what they watch and give them similar recommendations based on what they watch.

We would need to give our customers a disclaimer that we are collecting their data in order to give them the best possible recommendations. It should also be noted that we are in no manner selling their data to any third-party sources.

Data Load

We would utilize Amazon Web Server Glue in order to extract data from Flix2You's operational database and have it available for use in a dashboard, which would be hosted in Microsoft SQL Server, for Flix2You's executives.

Privacy Statement

At Flix2You, we are committed to protecting the privacy and security of our users. Here's how we collect, use, and safeguard your personal information.

COLLECTION OF PERSONAL INFORMATION:

We collect personal information from you when you sign up for our service, process a payment, or interact with us in any other way. This may include your name, email address, postal address, phone number, and payment information. We may also collect non-personal information such as your IP address, device type, and location.

USE OF PERSONAL INFORMATION:

We use your personal information to provide and improve our services, process payments, communicate with you, and comply with legal obligations. We may also use your information to personalize your user experience, recommend content, and conduct research and analysis.

DISCLOSURE OF PERSONAL INFORMATION:

We do not sell, rent, or disclose your personal information to third parties except as required by law or as necessary to provide our services. We may share your information with our affiliates, service providers, or partners who help us operate our business.

PROTECTION OF PERSONAL INFORMATION:

We take reasonable measures to protect your personal information from unauthorized access or disclosure. However, no method of transmission over the internet or electronic storage is 100% secure, and we cannot guarantee absolute security.

UPDATING AND DELETING PERSONAL INFORMATION:

You can update or delete your personal information by contacting us. We will make reasonable efforts to update or delete your information within a reasonable time frame.

CHANGES TO THIS PRIVACY STATEMENT:

Flix2You reserve the right to change this privacy statement at any time. Any material changes will be posted on our website, and we will notify you by email if necessary.

CONTACT INFORMATION:

If you have any questions or concerns about this privacy statement, please contact us at Support@Flix2You.com. We will do our best to address your concerns in a timely and professional manner.

Terms and Conditions for Flix2You

Welcome to Flix2You, an online streaming service where you can enjoy unlimited movies, TV shows, documentaries and much more. By accessing or using our website, you agree to comply with the following terms and conditions. Therefore, please read these Terms of Use and Privacy Policy carefully before using our services.

Eligibility

To use our services, you must be at least 18 years old or have reached the age at which you can lawfully enter and form contracts under applicable law. When using our services, you are representing and warranting that you are not under the age at which you can lawfully enter and form contracts under applicable law.

Compliance with Applicable Laws

You agree that you will not use the service for any illegal, prohibited, or unauthorized purpose. You must also comply with all domestic and international laws, rules, regulations and policies related to your use of our service.

Subscription Fee and Payment

By using our services, you agree to purchase a membership subscription and pay the applicable subscription fees. The membership subscription fees will be billed according to the plan you have selected, whether monthly or annually.

Your Account

When you subscribe to Flix2You, you agree to provide accurate and complete registration information. You are responsible for maintaining the confidentiality of your account login information and for all activities occurring under your account.

Content

Our website contains content that is owned or licensed by us, including movies, TV shows, documentaries, videos, audio, images, graphics, and software. You are permitted to stream and view content from our website for your personal non-commercial use only. You may create a personal watch list for your convenience.

Restrictions

You agree that you will not, directly or indirectly:

- (i) Copy, sell, rent, lease, distribute or otherwise transfer any part of the service or content, including any materials obtained through your use of the service.
- (ii) Modify, translate, reverse engineer, decompile or disassemble any part of the service, content or software.
- (iii) Impersonate or misrepresent your affiliation with any person or entity or falsely state or otherwise misrepresent your age, identity or affiliation.

(iv) Engage in any activity that is harmful, violent, obscene, pornographic or otherwise offensive.

Termination

We may, in our sole discretion, terminate or suspend your use of the service for any reason, including but not limited to, your breach of these terms and conditions. You agree that we shall not be liable to you or any third party for termination of your access to our website or service.

Disclaimer of Warranties

Our service is provided "as is" and "as available" without any warranty or representation of any kind, whether express or implied, including but not limited to, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement.

Limitation of Liability

In no event shall we be liable to you or any third party for any direct, indirect, incidental, special, consequential, or punitive damages arising out of your use of our website or service.

Governing Law

These terms and conditions shall be governed and construed in accordance with the laws of the United States of America without giving effect to any principles of conflicts of law.

Changes to Terms

We reserve the right to modify or update these terms and conditions at any time. You agree to review these terms and conditions periodically to be aware of any changes. Your continued use of the service after any modifications to these terms and conditions shall constitute your agreement to be bound by such modified terms and conditions.

Contact Us

If you have any questions about these terms and conditions, please contact us at support@flix2you.com.

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